Exercise 10

On applications of maxima and minima

From the school book | Understand | Apply | Higher Order Thinking Skills

1 *(1st Session 2024)* If $z = x$, find the smallest value of the expression $(z - \sqrt{x})$ « $\frac{-1}{4}$ »

2 The sum of two numbers is 30 and their product is as maximum as possible , find the two numbers. « 15 , 15 »

3 The sum of two numbers is 16 Find the two numbers if the sum of their squares is to be minimum. « 8 , 8 »

4 If the product of two positive numbers 36 , find the two numbers if their sum is to be minimum. « 6 , 6 »

5 The sum of two positive integers is 5 and the sum of the cube of the smaller number and double the square of the other is as minimum as possible , find the two numbers. « 2 , 3 »

6 *(2nd Session 2009)* The sum of three positive numbers is 36 and the greatest number of them is double the smallest of them. Find the three numbers such that their product is maximum. « 8 , 12 , 16 »

7 The sum of three positive number is 26 one of these numbers is three times one of the others Find the three numbers such that the sum of their squares has minimum value. « 4 , 12 , 10 »

8 *(2nd Session 2024)* Three positive numbers their sum equals 12 and the greatest number of them is three times the smallest of them , find these numbers such that their product is as maximum as possible. « 2 , 4 , 6 »

9 Find the positive number which if its multiplicative inverse is added to it , the sum is as minimum as possible. « 1 »

10 Find the maximum area of a rectangle - like piece of land that can be surrounded by a 120-metre fence. « 900 m^2 »

11 A rectangle is to be constructed using a piece of wire of length ℓ cm. Find its dimensions if its area is to be maximum. « $\frac{\ell}{4}$, $\frac{\ell}{4}$ »

12 If the perimeter of a circular sector is 30 cm. and its area is as maximum as possible , find the radius length of its circle. « 7.5 cm. »

13 If the perimeter of a circular sector is a constant , then prove that its area is to be maximum when measure of its central angle is 2 radians.

14 A circular sector - like coin whose area 16 cm^2. Find the radius length of the sector circle which makes its perimeter as minimum as possible. What is the measure of its angle then ? « 4 , 2^{rad} »

15 An open field is bounded by a straight river from one of its sides. Determine how to place a fence around the other sides of the rectangle-like piece of land to surround as maximum area as possible by a 800-metre fence. What is the area of the land then ?
« 200 , 400 m. , 80000 m^2 »

16 ***(2nd Session 2003)*** Find the dimensions of a rectangular piece of land. bounded by a river from one side and of area 2048 m^2 , such that the length of a fence surrounding its three remaining sides is minimum. « 32 , 64 m. »

17 ***(1st Session 2006)*** Find the points on the curve $x^2 - y^2 = 8$ such that their distance from the point (0 , 2) is minimum. « (3 , 1) , (– 3 , 1) »

18 Find the nearest point to point (0 , 5) and lies on the curve $y = \frac{1}{2}x^2 - 4$ « (– 4 , 4) , (4 , 4) »

19 Find the shortest distance between the straight line $x - 2y + 10 = 0$ and the curve $y^2 = 4x$ « $\frac{6\sqrt{5}}{5}$ length unit »

20 ***(2nd Session 2014)*** ABC is a right-angled triangle at B in which : AB + BC = 20 cm. Calculate the greatest possible area for this triangle « 50 cm^2 »

21 ***(1st Session 2007)*** The perimeter of an isosceles triangle is equal to 30 cm. Find the dimensions of the triangle such that its surface area is maximum. « 10 , 10 , 10 cm. »

22 If the hypotenuse length of a right-angled triangle equals 10 cm. , find the lengths of the two legs of the right angle when the area of the triangle is as maximum as possible.
« $5\sqrt{2}$, $5\sqrt{2}$ cm. »

23 ***(2nd Session 2015)*** Find the maximum value as possible of the expression : $M = 2x + y$ if x , y are lengths of the sides of the right angle in the right-angled triangle its hypotenuse length $\sqrt{5}$ cm. « 5 »

24 A right-angled triangle whose hypotenuse length is 30 cm. Find the length of the two legs of the right angle if the length of the altitude from the right angle on the hypotenuse is as maximum as possible. « $15\sqrt{2}$, $15\sqrt{2}$ cm. »

25 The area of a rectangle 16 cm^2 find its dimensions when the length of its diagonal is as minimum as possible. « 4 , 4 cm. »

26 Find the area of the largest rectangle that can be inscribed in a circle of radius 4 cm. « 32 cm^2 »

27 *(2nd Session 2006 , 2011)* Prove that the rectangle of greatest area which could be inscribed in a circle is a square and find the area of this square in terms of the radius length of this circle "r" « 2 r^2 square unit »

28 *(2nd Session 2000)* A thin metallic lamina in the form of a square , the length of whose side is 20 cm. A box without a lid in the form of a rectangular parallelepiped is to be made of this lamina by cutting equal squares of its corners , and turning up the sides. Find the length of the side of the removed square when the volume of the box is to be maximum. « $3\frac{1}{3}$ cm. »

29 A rectangle-like piece of cardboard paper whose two dimensions are 15 cm. and 24 cm. Congruent squares of side length x cm. are cut from its four corners , then the projected parts are folded upward to form a box without a cover , calculate the dimension of the box when its volume is as maximum as possible. « 18 , 9 , 3 cm. »

30 *(1st Session 2011)* A rectangular parallelepiped with a square base , the sum of all its edges equals 240 cm. Find the dimensions of the rectangular parallelepiped when its volume is to be maximum. « 20 , 20 , 20 cm. »

31 A rectangular parallelepiped the length of its base is twice its width. If the sum of its three dimensions 180 cm. Find these dimensions that make the volume of the rectangular parallelepiped is to be maximum. « 40 , 80 , 60 cm. »

32 *(2nd Session 2001)* A box in the form of a rectangular parallelepiped with a square base. Find its maximum volume given that its total surface area = 384 cm^2 « 512 cm^2 »

33 *(1st Session 2000 , 2nd Session 2004)* A rectangular parallelepiped has a volume of 576 cm^3 and the ratio between the two lengths of its base is 2 : 1 Find the dimensions of the parallelepiped that makes its total surface area minimum. « 6 , 12 , 8 cm. »

34 *(1st Session 2010)* If the sum of the lengths of the base radius of a right circular cylinder and its height is equal to 30 cm. , then find in terms of π the maximum volume of the cylinder. « 4000 π cm^3 »

35 If the total area of a right circular cylinder equals 150 π cm^2 Find the greatest volume of this cylinder. « 250 π cm^3 »

36 A cylinder shaped vessel opened from its upper base , its capacity 8000π cm^3 find its dimensions that makes its surface area is minimum as possible. « 20 , 20 cm. »

37 A closed cylindrical-like boxes are made of constant thickness to can beverages. The capacity of each box is k of volume units with minimum amount of materials. Find the ratio of the height (h) of the box to the radius length (r) of its base. « 2 : 1 »

38 The selling price of each unit of a product is $100 - 0.02\,x$ pounds. Where x is the number of units which are produced. If the cost price of x units is $40\,x + 1500$ pounds , find the number of units to be produced to satisfy the maximum profit. « 1500 units »

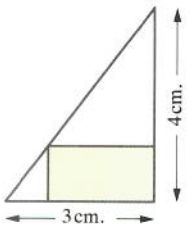
39 A point moves on the circle of radius length 10 cm. Find the distances of the point from the two ends of the diameter of the circle such that the sum of their distances is as maximum as possible. « $10\sqrt{2}$, $10\sqrt{2}$ cm. »

40 ***(1st Session 2013)*** Find the points on the curve of the function f where $f(x) = \frac{6}{x^2+3}$ at which the slope of the tangent is minimum and also the points at which the slope of the tangent is maximum. « $(1 , \frac{3}{2})$, $(-1 , \frac{3}{2})$ »

41 ***(1st Session 2009)*** If the length of two sides of a triangle are a , b where a , b are constants. Prove that the surface area of this triangle is maximum when the length of the third side is a diameter of the circle that passing through the vertices of the triangle.

42 ABCD is a rectangle in which AB = 4 cm. , BC = 9 cm. , draw straight line passing through the point C and cuts $\overrightarrow{AD}$ at E and cuts $\overrightarrow{AB}$ at O find the smallest area of the triangle AEO « 72 cm^2 »

43 Determine the dimensions of the rectangle of largest area that can be inscribed in the right-angled triangle shown in the figure.



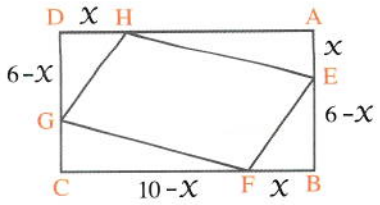
« 2 cm. , $\frac{3}{2}$ cm. »

44 In a perpendicular coordinate plane , $\overleftrightarrow{AB}$ is drawn to pass through the point C (3 , 2) and intersect the positive parts of the coordinate axes at point A and point B. Prove that the minimum area of triangle AOB equals 12 squared units where O is the origin point (0 , 0)

45 ABCD is a square whose side length 10 cm. and $M \in \overline{BC}$ where BM = x cm. and $N \in \overline{CD}$ where $CN = \frac{3}{2}\,x$, find the value of x which makes the area of Δ AMN as minimum as possible. « $\frac{10}{3}$ »

46 **In the opposite figure :**

ABCD is a rectangle :



(1) Prove that the figure EFGH is a parallelogram whose area $A = 2x^2 - 16x + 60$

(2) Find the possible minimum value of the area A

« 28 square unit »

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47 ABC is an isosceles triangle where AB = AC = 20 cm. , BC = 32 cm. , find the dimensions of the greatest rectangle can be drawn inside the triangle such that two vertices of the rectangle lie on the base of the triangle and the other two vertices lie on the other two sides of the triangle.

« 16 cm. , 6 cm. »

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48 ABC is an isosceles triangle , the length of its base $\overline{BC}$ equals 16 cm. and m ($\angle$ A) = 120° Find the area of the greatest rectangle can be drawn inside the triangle such that two vertices of the rectangle lie on the base of the triangle and the other two vertices of the rectangle lie on the other two sides of the triangle.

« $\frac{32\sqrt{3}}{3}$ cm.^{2} »

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49 A factory is producing electric appliances profits L.E. 30 in every appliance if it produces 50 appliances monthly. When the production increased than that , the profit in the appliance from all the produced appliances decreases by 50 piasters for every extra appliance produced. Find the number of appliances produced monthly if the profit is to be maximum.

« 55 appliance »

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50 🕮 The current intensity I (Ampere) in a circuit for the alternating current at any moment t (second) is given by the relation I = 2 cos t + 2 sin t. What is the maximum value of the current in this circuit ?

« $2\sqrt{2}$ Ampere »

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51 🕮 The volume of a bacterial culture placed in a food medium grows in regard to the relation $f(t) = 2000 + \frac{5000\,t}{100 + t^2}$ where time t is measures in hours , identify the maximum value of the culture volume.

« 2250 »

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52 A lake infected by bacteria has been treated by an antibacterial. If the number of bacteria z in 1 cm^3 after n day is given by the relation $z(n) = 20\left(\frac{n}{12} - \ln\left(\frac{n}{12}\right)\right) + 30$ such that $1 \leq n \leq 15$

(1) When the number of bacteria be minimum during this interval ?

(2) What is the least number of bacteria during this interval ?

« 12 days , 50 in each cm.^{3} »

53 A factory of iron production produces a quantity of good quality weights y tons and a quantity of less quality weights x tons. If the price of one ton from good quality is twice the price of one ton from the less quality , find the quantity which the factory must produce from each kind to get maximum income , given that $y = \frac{40 - 5x}{10 - x}$ where $x \neq 10$

« $x \approx 5.5$ tons , $y \approx 2.8$ tons »

54 A rectangle-like playground in which two opposite sides end in a semicircle outside the rectangle of a diameter length equal to the length of this side. If the perimeter of the playground is 400 metres , prove that the surface area of the playground is as maximum as possible when the ground is a circle-like , then find its radius length. « $\frac{200}{\pi}$ m. »

55 If the sum of the surface area of a sphere and the surface area of a right circular cylinder , whose radius is equal to the radius of the sphere is 250π cm^2 , then find the radius length of the sphere if the sum of their volumes is maximum. « 5 cm. »

56 A body consists of a cylinder above it there is a hemisphere such that its sharing the top surface with the cylinder , if the total surface area of the solid equals 2000π cm^2 and the volume of the solid is maximum , find the cylinder height and the length of its base radius.

« 20 , 20 cm. »

57 A covered tank for water is made up of a cuboid whose base is in the form of a square of side length $2x$ and height x and a right circular cylinder of diameter length $2x$ and height y is placed on the cuboid. If the total volume of the tank is 27 cm^3 , find the value of x which makes the surface area of the tank as minimum as possible. « $\frac{3}{2}$ »

58 *(2nd Session 2013)* A cylinder of volume V is cut from a solid sphere of radius r
Prove that the maximum volume of this cylinder is $\frac{4\pi r^3}{3\sqrt{3}}$

59 A right angled triangle whose hypotenuse length $\sqrt{3}$ cm. revolve around one of its right angled side to form a right circular cone, find the two side lengths of the triangle to get the greatest volume of the cone. « 1 cm. , $\sqrt{2}$ cm. »

60 ABC is a right angled triangle at B , AB + BC = 6 cm.
Find AB which makes volume of the solid formed by revolving the triangle ABC about $\overleftrightarrow{BC}$ a complete revolution as larger as possible. « 4 cm. »

61 *(2nd Session 2002)* A rectangle has one of its sides on the x-axis , the upper two vertices of the rectangle lie on the curve $y = 4 - x^2$ Determine the dimensions of the rectangle such that its area is maximum. « $\frac{4\sqrt{3}}{3}$, $\frac{8}{3}$ »

62 An isosceles triangle , its vertex is the origin and its base is a line segment parallels X-axis and lies above it and its endpoints lie on the curve $12\ y = 36 - X^2$, find the maximum possible area of this triangle. « $4\sqrt{3}$ square unit »

63 ***(1st Session 2012)*** Determine the maximum area of the trapezium ABCD in which $\overline{AB} \parallel \overline{CD}$, AB = AD = BC = 6 cm. « $27\sqrt{3}$ cm.^{2} »

64 An isosceles trapezoid has its smaller base of constant length (a) which is the same length as that of its two equal sides. Prove that its larger base equals (2a) when its area is maximum.

65 ***(1st Session 2002)*** ABCD is a trapezoid where $\overline{AD} \parallel \overline{BC}$, $\overline{AB} \perp \overline{BC}$, AB = 20 cm. , AD = 10 cm. , BC = 30 cm. Find the dimensions of the rectangle with the largest area which can be drawn inside the trapezoid. « 15 , 15 »

66 ***(1st Session 2016)*** Find the area of the largest trapezium that can be inscribed in a circle of radius 1 and whose base is a diameter of the circle. « $\frac{3\sqrt{3}}{4}$ square unit. »

67 If a trapezoid is drawn in a semi-circle such that is base is the diameter of the semi-circle , determine the measure of the angle of the trapezoid base such that its area is as maximum as possible. « 60° »

68 A rectangular parallelepiped box with an open top and a square base and volume 108 cubic metres is to be made. Find its dimensions if the total costs of making the box is to be minimum when the cost of one square metre is L.E. 12 and find this cost.

« 6 , 6 , 3 m. , 1296 L.E. »

69 🕮 A closed box-like tank whose capacity is 252 cubic metres and base is squared. If we want to coat the interior walls with an insulating material. The cost of the bottom is 50 L.E. per square metre , the cost of cover is 20 L.E. per square metre and the cost of the walls is 30 L.E. per square metre , find the dimensions of the box which makes the cost as minimum as possible. « 6 , 6 , 7 m. »

70 🕮 An open tank (without lid) of squared base , vertical sides and holds a certain amount of water , prove that the cost of coating the tank from interior by a regular insulating material is as minimum as possible if its depth equals half the side length of the base.

71 A semi-spherical top vertical cylinder-like silo is needed to be constructed where it holds 108π m^3 of wheat to store (let the wheat be stored in the cylindric part only but not the top) , and the cost of the surface unit area from its top is twice the cost of the unit area of the lateral wall. What are the dimensions of the silo which make the cost minimum ?

« 3 , 12 m. »

72 ***(1st Session 2003)*** The sum of the circumference of a circle and the perimeter of a square is 80 cm. Prove that when the sum of the areas of both figures is minimum , then the diameter of the circle equals to the side length of the square.

73 A wire of length 34 cm. is divided into two parts. A square is to be constructed from one part and a circle is to be constructed from the second part. Find the length of each part if the sum of the area of the two figures is to be minimum.

« $\frac{34\pi}{\pi+4}$, $\frac{136}{\pi+4}$ »

74 ***(1st Session 2001)*** A wire of length 68 cm. is cut into two pieces. The first is bent to form a rectangle of width x cm. and of length twice its width , while the second is bent to form a square. Find the value of x such that the sum of the rectangle and square areas is minimum.

« 6 cm. »

75 A window is formed from a rectangle headed by an isosceles triangle its base is coincide with one of the sides of the rectangle , if the height of the triangle = $\frac{3}{8}$ the length of its base and the perimeter of the window is 120 cm. Find the two dimensions of the rectangle which makes the area of the window is to be maximum.

« 32 , 24 cm. »

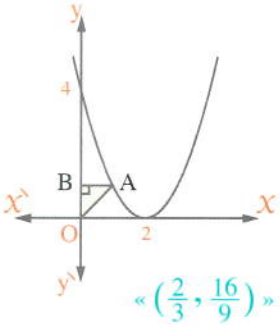
76 A window is formed from a rectangle headed by a semi-circle whose diameter is coincide with one of the sides of the rectangle. If the perimeter of the window equals 6 metres , find the radius length of the semi-circle so that the area of the window is to be maximum.

« $\frac{6}{4+\pi}$ »

77 **In the opposite figure :**

If the point A $\in$ the curve of the quadratic function $y = (x-2)^2$, $\overline{AB}$ // x-axis.

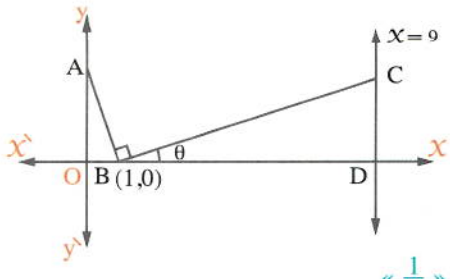
Find A at which the area of triangle OAB as large as possible.



« $(\frac{2}{3}, \frac{16}{9})$ »

78 **In the opposite figure :**

Find the value of tan θ which makes (AB + BC) as small as possible.



« $\frac{1}{2}$ »

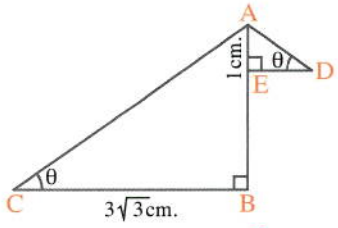
79 **In the opposite figure :**

If $\overline{AB} \perp \overline{BC}$, $\overline{DE} \perp \overline{AB}$,

AE = 1 cm. , BC = $3\sqrt{3}$ cm. , m (∠ D) = m (∠ C) = θ

Find the length of $\overline{BE}$ which makes

$(AD + AC)^2$ as small as possible.



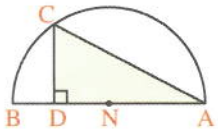
« 2 cm. »

80 **In the opposite figure :**

$\overline{AB}$ is a diameter in a semi-circle ,

AB = 16 cm.

Find the greatest area of ΔADC



« $24\sqrt{3}$ cm^2. »

81 **Choose the correct answer from the given ones :**

(1) The greatest value of the expression $8x - x^2$ where $x \in \mathbb{R}$ is .....

(a) 8 (b) 12 (c) 16 (d) 20

(2) *(2nd Session 2023)* If $y = 3x - x^4$, then the maximum value for the expression : $(y + x)$ equals .....

(a) – 4 (b) – 2 (c) 2 (d) 3

(3) If $x + y = 13$ where x and y are positive numbers , then the value of x which makes the expression $(2x + 3y + xy)$ has a maximum value is .....

(a) 2 (b) 4 (c) 6 (d) 8

(4) *(1st Session 2021)* If $x \in \mathbb{R}^+$, then the smallest value of $\sqrt{x} + \frac{1}{x}$ equals .....

(a) $2\sqrt[3]{4}$ (b) $\frac{\sqrt[3]{4}}{2}$ (c) $\frac{3}{\sqrt[3]{4}}$ (d) $\frac{\sqrt[3]{4}}{3}$

(5) If the perimeter of a rectangle 14 cm. , then the greatest area for it equals ..... cm^2

(a) 10 (b) 12 (c) 12.25 (d) 49

(6) If $x < 0$, then the greatest value of the expression $x + \frac{1}{x}$ equals .....

(a) – 1 (b) – 2 (c) – 3 (d) otherwise.

(7) The maximum value of the expression $(\sin x + \sqrt{3}\cos x)$ to be when $x =$ .....

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) zero

(8) The range of the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ where $f(x) = \sin x + \cos x$ equals .....

(a) $[-\sqrt{2}, \sqrt{2}]$ (b) $[-1, 1]$ (c) $[0, 1]$ (d) $[0, \sqrt{2}]$

(9) The minimum value of the function $(x \ln x)$ equals .....

(a) e (b) $\frac{1}{e}$ (c) $\frac{-1}{e}$ (d) $-e$

(10) The rectangles which drawn inside circle with radius length (r) , then the dimensions of the rectangle of the maximum area are .....

(a) $\sqrt{2}$ r , $\sqrt{2}$ r (b) r , $\sqrt{2}$ r (c) 2 r , $\sqrt{2}$ r (d) 2 r , 2 r

(11) If the perimeter of circular sector is constant p , then the area be maximum value when r = ...

(a) $\frac{p}{2}$ (b) $\frac{2}{\sqrt{p}}$ (c) $\sqrt{p}$ (d) $\frac{p}{4}$

(12) The height of circular cylinder which has maximum volume inside sphere of radius length (r) equals .....

(a) $\frac{2r}{\sqrt{5}}$ (b) $\frac{2r}{\sqrt{3}}$ (c) 2 r (d) $2\sqrt{3}$ r

(13) The height of a cone which has maximum volume inside a sphere with radius length r equals .....

(a) $\frac{4r}{5}$ (b) $\frac{4r}{3}$ (c) $\frac{2r}{3}$ (d) $\frac{8r}{3}$

(14) Height of a right cone we can put it inside a sphere with radius length 9 cm. , such that its volume maximum value , equals ..... cm.

(a) 7 (b) 12 (c) 8 (d) -1

(15) The minimum distance between the origin point and the curve $x = 2\sin t - \sin 2t$, $y = 2\cos t - \cos 2t$, equals ..... length unit.

(a) 1 (b) 2 (c) 3 (d) 4

(16) *(Speciman 2025)* The shortest distance between the origin point and the curve $x y^2 = 1$ is at the point ..... that lies on the curve.

(a) (1 , 1) (b) $(2, \frac{1}{4})$ (c) $(2^{\frac{1}{6}}, 2^{\frac{-1}{3}})$ (d) $(2^{\frac{-1}{3}}, 2^{\frac{1}{6}})$

(17) *(Speciman 2025)* A player jumps with pole vault following the track curve :

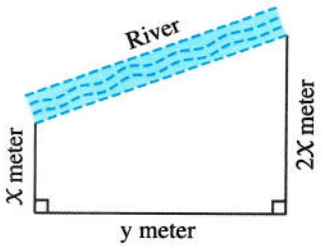
$y = \frac{-1}{10000} x^4 + 7$, then the maximum height the player reach to it is ..... unit length.

(a) 0.007 (b) 7 (c) 0.0028 (d) 0.7

(18) If L , M are the two roots of the equation $x^2 - (k-1)x + k + 1 = 0$, then k which makes the expression $L^2 + M^2$ as minimum as possible equals .....

(a) 2 (b) 3 (c) 1.5 (d) – 1.5

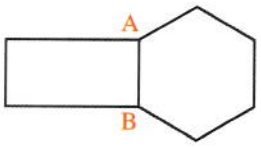
(19) *(1st Session 2025)* The figure shows an open piece of land in shape of a right angled trapezium bounded by a straight river from one of its sides as the figure , it is wanted to surround it by a fence of length 120 meter around the other three sides , then the maximum area of the piece of land = ... squared meter.



(a) 1200 (b) 1800 (c) 3510 (d) 3600

(20) *(Speciman 2025)* **In the opposite figure :**

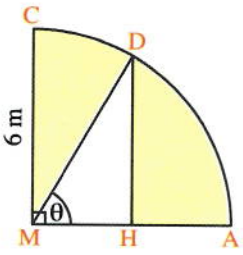
Two gardens were separated , one in the shape of a rectangle which planted by apples and the other in the shape of regular hexagon which planted by roses.



The two gardens except $\overline{AB}$ are surrounded by **double wire fence**.
If the length of the wire is 120 meters , then the maximum planted area by apples is .....

(a) 60 (b) 75 (c) 120 (d) 150

(21) *(Speciman 2025)* If the opposite figure represents a quarter of a circle of center M , then the minimum area of the shaded part of the figure equals .....

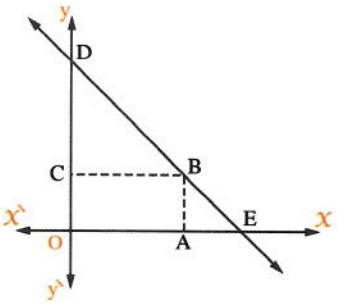


(a) $9\pi + 9$ (b) $6\pi + 6$

(c) $6\pi - 6$ (d) $9\pi - 9$

(22) In the opposite figure :

If the equation of the straight line $\overleftrightarrow{DE}$ is $y = 3 - x$, then the greatest area of the rectangle ABCO = ... square unit.



(a) $\frac{9}{8}$ (b) $\frac{9}{2}$

(c) $\frac{9}{4}$ (d) $\frac{3}{2}$

(23) In the opposite figure :

A quarter of a circle of centre (O)

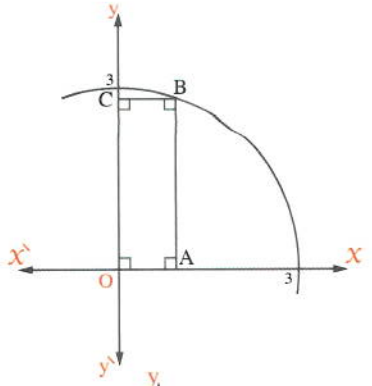
, B $\in$ the curve of the circle $x^2 + y^2 = 9$

, then the greatest area of the rectangle

ABCO = ... square unit.

(a) 4.5 (b) 9

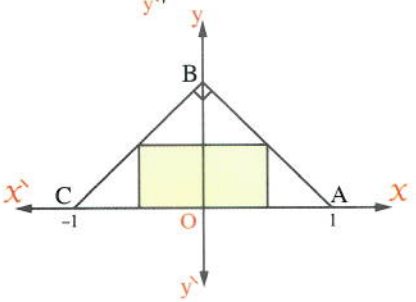
(c) $\frac{3\sqrt{2}}{2}$ (d) 27



(24) The opposite figure represents a rectangle inscribed in an isosceles right-angled triangle AC = 2 length units then the greatest area of the rectangle equals ..... square unit(s)

(a) $\frac{1}{2}$ (b) 1

(c) $1\frac{1}{2}$ (d) 2

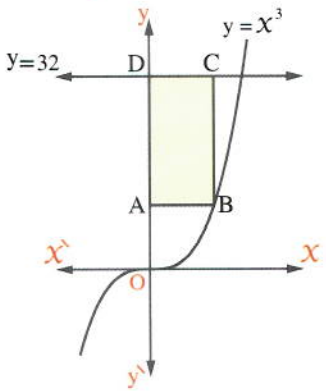


(25) In the opposite figure :

$f(x) = x^3$, then the maximum area of the rectangle ABCD equals ..... square unit.

(a) 2 (b) 8

(c) 48 (d) 24



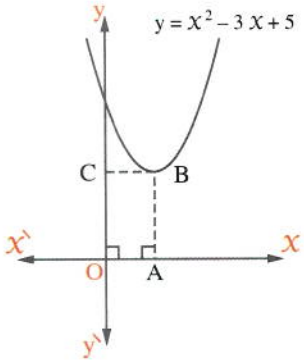
(26) In the opposite figure :

B $\in$ the curve $y = x^2 - 3x + 5$

then the minimum value of the perimeter of the rectangle OABC equals ..... length unit.

(a) 3 (b) 8

(c) 12 (d) 16



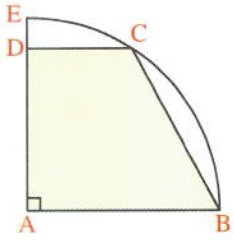
(27) In the opposite figure :

A quarter circle with radius 4 cm.

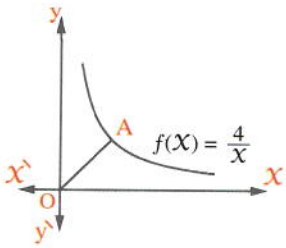
, the greatest area of the trapezium ABCD equals ..... cm^2

(a) 6 (b) $6\sqrt{3}$

(c) 12 (d) $12\sqrt{3}$



(28) In the opposite figure :

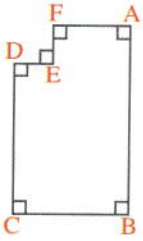


The least length of the line segment $\overline{OA}$ = ... length unit.

(a) $\sqrt{2}$ (b) 2

(c) $2\sqrt{2}$ (d) 4

(29) In the opposite figure :



If CD = 2 AF and FE = ED

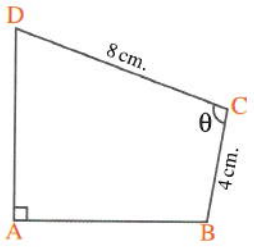
and the perimeter of the figure ABCDEF = 40 cm.

, then the maximum area of the figure ABCDEF equals ..... cm^2

(a) 90 (b) 95

(c) 100 (d) 105

(30) In the opposite figure :



ABCD is a quadrilateral m (∠ A) = 90° , m (∠ C) = θ

AB = AD , BC = 4 cm. , CD = 8 cm.

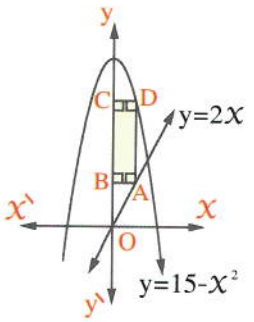
If the area of the figure ABCD is maximum

, then θ = ...

(a) 105° (b) 120°

(c) 135° (d) 150°

(31) In the opposite figure :

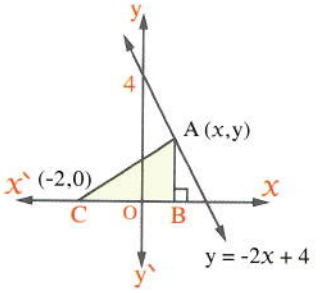


The greatest area of the rectangle ABCD equals ..... square units.

(a) $\frac{400}{27}$ (b) $\frac{401}{27}$

(c) $\frac{403}{27}$ (d) $\frac{404}{27}$

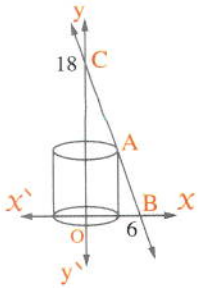
(32) *(Trial 2021)* In the opposite figure :



If the point A (X , y) moves on the straight line $y = -2X + 4$ where $X \in [0 , 1]$ and the point B is the projection of A on the X-axis , c (– 2 , 0) , then the least possible value of area of Δ ABC equals ..... square unit.

(a) 2 (b) 3 (c) 4 (d) 5

(33) *(Speciman 2025)* **In the opposite figure :**

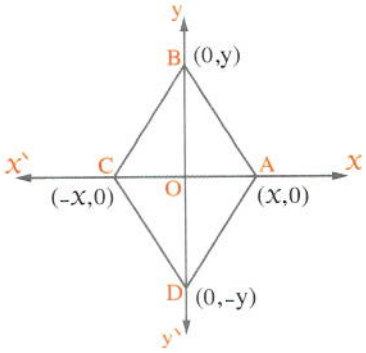


The greatest volume of right circular cylinder is ..... cubic unit.

(a) 72π (b) 96π

(c) 120π (d) 144π

(34) *(2nd Session 2021)* **In the opposite figure :**

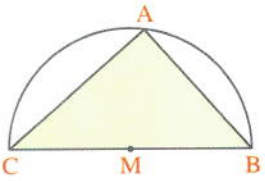


If AB = 5 cm. , then the area of the figure ABCD is maximum when .....

(a) $x = y$ (b) $x = 2y$

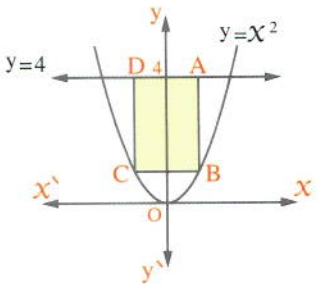
(c) $y = 2x$ (d) $3x = 4y$

(35) *(2nd Session 2022)* The opposite figure represents a semicircle M with radius length r length units and $\overline{BC}$ is its diameter, then the greatest area of the triangle ABC = ... square units.



(a) $\frac{1}{4}\pi r^2$ (b) $2r^2$ (c) r^2 (d) $\frac{1}{2}\pi r$

(36) *(1st Session 2022)* **In the opposite figure :**

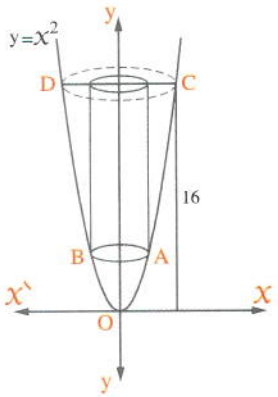


The greatest possible area of the rectangle ABCD = ... area units.

(a) $\frac{16\sqrt{3}}{3}$ (b) $\frac{32\sqrt{3}}{3}$

(c) $\frac{32\sqrt{3}}{9}$ (d) $\frac{16\sqrt{3}}{9}$

(37) *(Speciman 2025)* **In the opposite figure :**

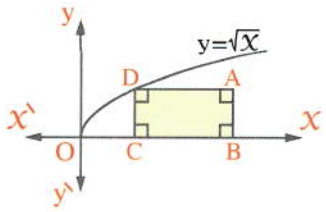


If the curve $y = x^2$ revolving a complete rotation about y-axis and a right circular cylinder is put inside it , its diameter $\overline{AB}$ parallel to x-axis , then the greatest volume of the right circular cylinder = ... cm^3.

(a) 14π (b) 36π

(c) 48π (d) 64π

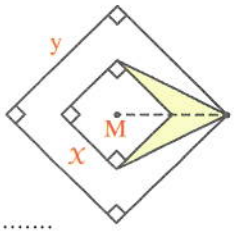
(38) In the opposite figure :



If B = (12 , 0) , then the greatest volume of the solid generated by revolving the shaded region a complete revolution about X-axis = ... cubic unit.

(a) 48π (b) 36π (c) 24π (d) 18π

(39) *(2nd Session 2021)* In the opposite figure :

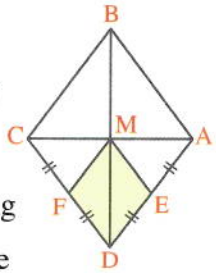


Two squares having the same center and their sides lengths are 1 cm. and 4 cm. , if side length of the first square increases by rate 1 cm./sec. and side length of the second square decreases by rate $\frac{1}{2}$ cm./sec. , then the area of the shaded region after $\frac{1}{2}$ second .....

(a) stop increasing instantly. (b) stop decreasing instantly.

(c) increases at rate $\frac{1}{4}$ cm^2/sec. (d) decreases at rate $\frac{1}{4}$ cm^2/sec.

(40) *(1st Session 2022)* In the opposite figure :



ABCD is a rhombus, M is the intersection point of its diagonals , E and F are the midpoints of $\overline{AD}$ and $\overline{CD}$ respectively The lengths of the two diagonals $\overline{AC}$ and $\overline{BD}$ are 6 cm. and 8 cm. respectively.

If AC started increasing by rate 1 cm./sec. while BD started decreasing by rate $\frac{1}{2}$ cm./sec. , then the surface area of the region MEDF at the end of the fifth second will .....

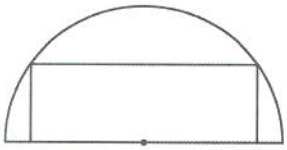
(a) stop instantly increasing. (b) stop instantly decreasing.

(c) increase at rate 2 cm^2/sec. (d) decrease at rate 2 cm^2/sec.

82 If a straight line passes through a fixed point (a , b) lies on the 1st quadrant to intersect the two positive parts of coordinate axes , prove that the minimum sum of lengths of axes interception is $(\sqrt{a}+\sqrt{b})^2$

83 *(2nd Session 2008)* ABC is a right-angled triangle at B and AB = 6 cm. , BC = 8 cm. , D is the midpoint of $\overline{AB}$, $E \in \overline{BC}$, $N \in \overline{AC}$ Find the surface area of the shape ADEN when BE × EC is maximum and $(AN)^2 + (NC)^2$ is minimum. « 12 cm.^{2} »

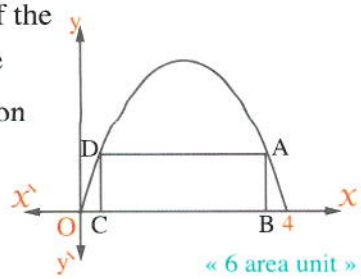
84 *(2nd Session 2010)* **In the opposite figure :**



A rectangle is drawn inside the surface of semi-circle with radius length 4 cm. Find the dimensions of this rectangle when its area is maximum.

« $4\sqrt{2}$, $2\sqrt{2}$ cm. »

85 *(2nd Session 2012)* The opposite figure represents the curve of the function f where $f(x) = 4 - (x-2)^2$, $0 \leq x \leq 4$ the rectangle ABCD is drawn such that A , D lie on this curve of the function f and $\overline{BC} \subset \overrightarrow{OX}$ Find the area of the rectangle ABCD at which the rectangle has greatest perimeter.

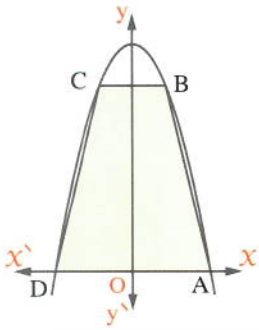


« 6 area unit »

86 The opposite figure represents the curve of the function $y = 9 - x^2$

Find the maximum area of the trapezium ABCD

« 32 square unit »

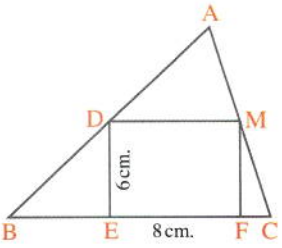


87 Find the two dimensions of a rectangle which has the biggest area and it can be drawn inside a triangle whose base length is 16 cm. and its height is 12 cm. such that one of its sides lies on the base of the triangle and the two vertices of the opposite side lie on the two other sides of the triangle.

« 6 , 8 cm. »

88 *(1st Session 2008)* **In the opposite figure :**

ABC is a scalene triangle , DEFM is a rectangle in which EF = 8 cm. , DE = 6 cm. , $D \in \overline{AB}$, $M \in \overline{AC}$, $\overline{EF} \subset \overline{BC}$ Find the minimum value of the surface area of the triangle ABC

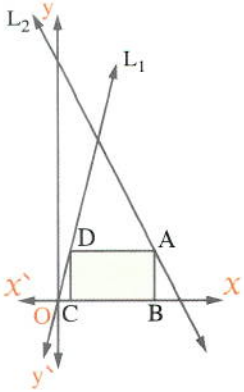


« 96 cm.^{2} »

89 **In the opposite figure :**

Two straight lines $L_1 : y = 4x$, $L_2 : y = 18 - 2x$

Find the greatest area of the rectangle ABCD



« 27 square units »

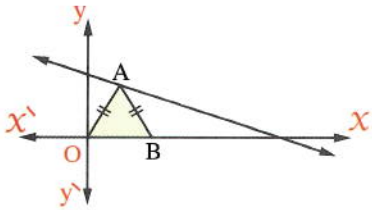
90 *(Speciman 2025)* A piece of wire in the shape of a circle of radius length 12 cm. , it is wanted to divide it into two pieces to form a circle from each piece. Find the length of each piece that make the sum of areas of the two circles to be minimum.

« 12π cm. , 12π cm. »

Higher skills

91 **Choose the correct answer from the given :**

(1) In the opposite figure :

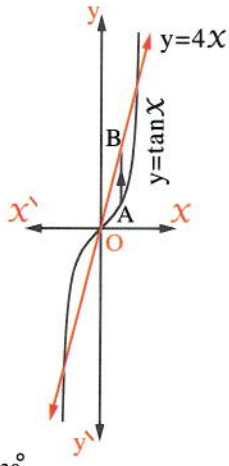


A $\in$ the straight line $x + 3y = 6$ and lies in the first quadrant , the greatest area of the isosceles triangle ABO = ... square units.

(a) 2 (b) 3

(c) 4 (d) 6

(2) In the opposite figure :

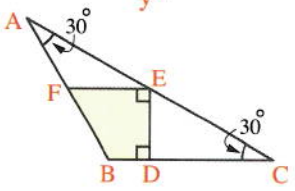


If $\overleftrightarrow{AB}$ // the y-axis , then the greatest value of length of $\overline{AB}$ = ... cm.

(a) $\frac{4\pi - 3\sqrt{3}}{3}$ (b) $\frac{4\pi + 3\sqrt{3}}{3}$

(c) $\frac{2\pi - 3\sqrt{6}}{3}$ (d) $\frac{2\pi + 3\sqrt{3}}{3}$

(3) In the opposite figure :



ABC is an isosceles triangle AB = BC = 10 cm. , m ($\angle$ C) = 30° , then the greatest area of the right trapezium BDEF equals ..... cm^2.

(a) $8\sqrt{3}$ (b) $10\sqrt{3}$ (c) $12\sqrt{3}$ (d) $15\sqrt{3}$

(4) An advertising poster of area 2 m^2. is required to be made. There is a fram around its printed area such that there is 21 cm. above and below the printed area and 14 cm. from each side left and right , then the dimensions of the poster that makes the printed area is maximum are ..... m.

(a) $\sqrt{2}$, $\sqrt{2}$ (b) $\sqrt{3}$, $\frac{2\sqrt{3}}{3}$ (c) $\sqrt{5}$, $\frac{2\sqrt{5}}{5}$ (d) $\frac{1}{2}$, 4

92 A rectangle like poster contains 800 cm^2 of the printed material where the widths of both lower and upper margins are 10 cm. and the two side margins are 5 cm.

What are the two dimensions of the poster which make its area as minimum as possible.

« 60 , 30 cm. »

---

93 📖 $\overline{AB}$ is a diameter in a circle of a radius length r , two tangent are drawn to the circle at A and B , from point E on the circle another tangent is drawn to the circle to intersect the two previous tangents at D and C respectively , prove that the minimum area of the trapezium ABCD equals 2 r^2 squared unit.

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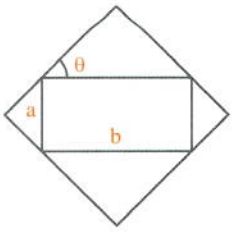
94 ABC is an equilateral triangle whose side length 2 ℓ , E is the midpoint of $\overline{BC}$, $D \in \overline{AB}$, $F \in \overline{AC}$ where $\overline{DF} \parallel \overline{BC}$

Prove that : The greater area of the triangle DFE = $\frac{1}{4}$ the area of Δ ABC

---

95 ***(1st Session 2004)*** **In the opposite figure :**

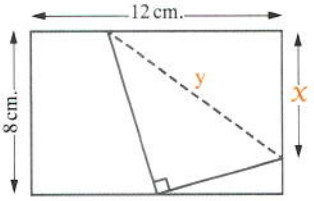
Find the largest surface area of the rectangle that can be drawn outside the rectangle of side lengths the two constants a and b



« $\frac{1}{2}(a+b)^2$ square unit »

---

96 The top right corner of a rectangular piece of paper whose dimension 8 cm. , 12 cm. is folded to align the lower edge as shown in the figure. Find the value of x which makes y as minimum as possible.



« 6 cm. »

---

97 Let A (0 , 9) , B (0 , 4) , $C \in \overrightarrow{Ox}$, find the coordinates of C which make the measure of $\angle$ ACB is as great as possible. « (6 , 0) »

---

98 ***(1st Session 2014)*** Find the area of the largest isosceles triangle that can be inscribed in a circle of radius 15 cm. « 292.28 cm^2 »

---

99 📖 A 2 metre wall and it is 2 metres away from a house , find the minimum length of the ladder that can be used to connect the ground and the house resting on the wall.

« $4\sqrt{2}$ m. »

100 A rectangle is drawn such that two adjacent vertices of the rectangle lie on the curve $y = x^2 - 12$ and the other two vertices lie on the curve $y = 12 - x^2$, find the maximum area of this rectangle.

« 64 square unit »

---

101 A cube with side length $\sqrt[3]{2}$ L is melted and changed to right prism whose base is an equilateral triangle. Prove that the total surface area of the prism is to be minimum when the length of the side of the base is 2 L

---

102 Find the height of a right circular cylinder that can be put in a hollow sphere of interior radius length 10 cm. when the lateral area of the cylinder is as maximum as possible.

« $10\sqrt{2}$ cm. »

---

103 Find the dimensions of the right circular cylinder that can be drawn inside a right circular cone of height 24 cm. and radius length of its base 10 cm. when the volume be maximum.

« $\frac{20}{3}$, 8 cm. »

---

104 Prove that the volume of the greatest cone can be drawn inside a sphere of radius length (a) equals $\frac{8}{27}$ of the volume of the sphere.

---

105 A man in a rowboat at a point C 5 km. from the nearest point A on a straight shoreline wishes to reach a point B 6 km. from a farther down the shore. If the man can row at a rate of 2 km./hr. and walk at a rate of 4 km./hr. , where should he land in order to reach B in the least time.

« $\frac{5\sqrt{3}}{3}$ metre from A »